We create chemistry

Safety Data Sheet
RETINOL 50 C

1. Identification

Product identifier used on the label

RETINOL 50 C

Recommended use of the chemical and restriction on use
Recommended use*: Chemical
Unsuitable for use: Not intended for sale to or use by the general public.

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA

Telephone: +1 973 245-6000

Emergency telephone number
24 Hour Emergency Response Information
CHEMTREC: 1-800-424-9300
BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification
Synonyms: Retinol (and) Polysorbate 20

2. Hazards Identification

According to Regulation 2024 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Skin Irrit.	2	Skin irritation
Skin Sens.	1	Skin sensitization
Aquatic Acute	2	Hazardous to the aquatic environment - acute
Carc.	2	Carcinogenicity
Repr.	1B (unborn child)	Reproductive toxicity
Aquatic Chronic	2	Hazardous to the aquatic environment - chronic

Label elements

Pictogram:



Signal Word:
Danger

Hazard Statement:

H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H351	Suspected of causing cancer.
H360	May damage the unborn child.
H401	Toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P201	Obtain special instructions before use.
P261	Avoid breathing mist or vapour or spray.
P273	Avoid release to the environment.
P202	Do not handle until all safety precautions have been read and understood.
P272	Contaminated work clothing should not be allowed out of the workplace.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P308 + P313	IF exposed or concerned: Get medical attention.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P333 + P313	If skin irritation or rash occurs: Get medical attention.
P332 + P313	If skin irritation occurs: Get medical attention.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P391	Collect spillage.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents/container in accordance with local regulations.

Hazards not otherwise classified

No specific dangers known, if the regulations/notes for storage and handling are considered.

3. Composition / Information on Ingredients

According to Regulation 2024 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

BHA

CAS Number: 25013-16-5

Content (W/W): >= 0.3 - < 1.0%
Synonym: 1,1-(Dimethylethyl)-4-methoxyphenol; Butylhydroxyanisole

Retinol

CAS Number: 68-26-8
Content (W/W): >= 25.0 - < 50.0%
Synonym: (All-E)-3,7-Dimethyl-9-(2,6,6-trimethyl-1-cyclohexene-1-yl)-2,4,6,8-nonatetraen-1-ol; Vitamin A

BHT

CAS Number: 128-37-0
Content (W/W): >= 3.0 - < 5.0%
Synonym: 2,6-Bis(1,1-dimethylethyl)-4-methylphenol; BHT, Butylated hydroxytoluene, 2,6-Di-tert-butyl-p-cresol

4. First-Aid Measures

Description of first aid measures

General advice:
First aid personnel should pay attention to their own safety. Remove contaminated clothing.

If adverse health effects develop seek medical attention.

If inhaled:
After inhalation of vapor: fresh air.

If on skin:
Wash thoroughly with soap and water

Do not rub affected parts of the body. Remove contaminated clothing. If irritation develops, seek medical attention.

If in eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open. Seek medical attention.

If swallowed:
Rinse mouth and then drink 200-300 ml of water.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Information on: Retinol
Symptoms: Overexposure may cause:, Eye irritation, skin irritation, erythema, nausea, headache, vomiting, dizziness, diarrhea, abdominal cramps
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Hazards: Allergic skin reaction, bronchospasms and anaphylactic shock possible.

Indication of any immediate medical attention and special treatment needed

Note to physician
Treatment: Symptomatic treatment (decontamination, vital functions).

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, carbon dioxide, dry powder, foam

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
harmful vapours
Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Protective equipment for fire-fighting:
Firefighters should be equipped with self-contained breathing apparatus and turn-out gear.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

Do not breathe vapour/spray.

Use personal protective clothing.

Environmental precautions

Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Vacuum up spilled product.
For residues: Contain with absorbent material (e.g. sand, silica gel, acid binder, general purpose binder, sawdust).
Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Precautions for safe handling

Processing machines must be fitted with local exhaust ventilation. Avoid contact with the skin, eyes and clothing.

Protection against fire and explosion:

Take precautionary measures against static discharges. Avoid all sources of ignition: heat, sparks, open flame.

Temperature class: T3 (Autoignition temperature >200 °C).

Conditions for safe storage, including any incompatibilities

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Aluminium

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place. Keep under inert gas. Protect contents from the effects of light. Keep at temperature not exceeding 20 °C.

Storage stability:
Storage temperature: 2 - 8 °C
Protect from temperatures below: 0 °C
Characteristics of the product are irreversibly changed below the limit temperature.
Protect from temperatures above: 50 °C
Damage by exceeding the maximum temperature is not reversible. The packed product must be protected against exceeding the indicated temperature.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

BHT	ACGIH, US:	TWA value 2 mg/m3 Inhalable fraction and vapor ;

Advice on system design:
No further information, see section 7.

Personal protective equipment

Respiratory protection:
Wear a NIOSH-certified (or equivalent) organic vapour/particulate respirator as needed.

Hand protection:
Chemical resistant protective gloves

Eye protection:
Safety glasses with side-shields.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures:
Under no circumstances should the product come into contact with the skin of pregnant women or be inhaled by them. Handle in accordance with good industrial hygiene and safety practice. No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Physical state:	liquid	
Form:	highly viscous, liquid	
Odour:	characteristic	
Odour threshold:	not determined	
Colour:	yellow to amber	
pH value:	substance/mixture is non-soluble (in water), not relevant	
Freezing point:	-35 °C	
Boiling point:	134 °C	(OECD Guideline 103)
Flash point:	202 °C	(ISO 2592)
Flammability:	not highly flammable	
Flammability of Aerosol Products:	not applicable, the product does not form flammable aerosoles	
Lower explosion limit:	For liquids not relevant for classification and labelling.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Autoignition:	286 °C	(DIN EN 14522)
Vapour pressure:	12 hPa (29 °C) approx. 44 hPa (50 °C)	
Density:	1.025 g/cm3 (20 °C)	(ISO 2811-3)
Relative vapour density:	not determined	
Partitioning coefficient n-octanol/water (log Pow):	not determined	
Self-ignition temperature:	not self-igniting	
Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.	
Viscosity, dynamic:	not determined	
Viscosity, kinematic:	not determined	
Solubility in water:	of low solubility	
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	
Other Information:	If necessary, information on other physical and chemical parameters is indicated in this section. No further information available.	

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Oxidizing properties:
Based on its structural properties the product is not classified as oxidizing.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid
Avoid prolonged exposure to extreme heat. Avoid contact with air.

Incompatible materials
oxidizing agents

Hazardous decomposition products

Decomposition products:
Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

Thermal decomposition:
No decomposition if stored and handled as prescribed/indicated.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity
Assessment of acute toxicity: Virtually nontoxic after a single ingestion.

Oral
Type of value: LD50
Species: rat (male/female)
Value: > 2,000 mg/kg (BASF-Test)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Dermal
Type of value: ATE
Value: > 5,000 mg/kg

Assessment other acute effects
Assessment of STOT single:
Based on available data, the classification criteria are not met.

Irritation / corrosion
Assessment of irritating effects: Not irritating to the eyes. Irritating to skin.

Information on: Retinol
Assessment of irritating effects: Skin contact causes irritation. Eye contact causes irritation.

Skin
Result: Irritant.
Method: Epiderm Corrosivity Test

Eye
Species: rabbit
Result: non-irritant
Method: OECD Guideline 405

Sensitization
Assessment of sensitization: May cause sensitization by skin contact.

Information on: Retinol
Assessment of sensitization:
Sensitization after skin contact possible.

Open epicutaneous test (OET)
Species: guinea pig
Result: skin sensitizing
Method: OECD Guideline 406

Buehler test
Species: guinea pig
Result: skin sensitizing
Method: OECD Guideline 406

Aspiration Hazard
No aspiration hazard expected.

Chronic Toxicity/Effects

Repeated dose toxicity
Assessment of repeated dose toxicity: Overexposure may cause blood abnormalities. May cause joint/bone pain, fatigue, depression, skin rash, liver and spleen abnormalities, bone changes and CNS effects.
Repeated exposure to large quantities may affect certain organs.

Information on: Retinol
Assessment of repeated dose toxicity: Repeated exposure to large quantities may affect certain organs.
Overexposure may cause blood abnormalities. May cause joint/bone pain, fatigue, depression, skin rash, liver and spleen abnormalities, bone changes and CNS effects.

Genetic toxicity
Assessment of mutagenicity: The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture.
Genetic toxicity in vitro: Modified OECD Guideline 471 Ames-test S. typhimurium, E. coli:with and without metabolic activation negative

Carcinogenicity

Assessment of carcinogenicity: A carcinogenic potential cannot be excluded after prolonged exposure to severely irritating concentrations. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: BHA
Assessment of carcinogenicity: Indication of possible carcinogenic effect in animal tests. IARC (International Agency for Research on Cancer) has classified this substance as group 2B (The agent is possibly carcinogenic to humans).

Reproductive toxicity
Assessment of reproduction toxicity: The chemical structure does not suggest a specific alert for such an effect.

Information on: Retinol
Assessment of reproduction toxicity: No reliable data are available concerning reproduction toxicity.

Teratogenicity
Assessment of teratogenicity: The substance caused malformations/developmental toxicity in laboratory animals. Literature data.

Information on: Retinol
Assessment of teratogenicity: May cause harm to the unborn child.

Experiences in humans
There are indications of coherence between the incorporation of high doses during pregnancy and deformations at humans (newborn child).

Other Information
Females of childbearing age should not come into contact with the product.

12. Ecological Information

Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. Very toxic (acute effect) to aquatic organisms. May cause long-term adverse effects in the aquatic environment.

Aquatic invertebrates
EC50 (48 h) 0.018 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The statement of the toxic effect relates to the analytically determined concentration.

Aquatic toxicity

Information on: BHT
Assessment of aquatic toxicity:
Very toxic (acute effect) to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: Retinol
Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms
OECD Guideline 209 aerobic
activated sludge, domestic/EC20 (3 h): > 1,000 mg/l
The details of the toxic effect relate to the nominal concentration.

Persistence and degradability

Elimination information

40 - 50 % BOD of the ThOD (28 d) (OECD 301F; ISO 9408; 92/69/EWG, C.4-D) (aerobic, activated sludge, domestic)

Bioaccumulative potential

Assessment bioaccumulation potential
No data available.

Mobility in soil

Assessment transport between environmental compartments
not applicable

13. Disposal considerations

Waste disposal of substance:
Dispose of in accordance with national, state and local regulations. It is the waste generator's responsibility to determine if a particular waste is hazardous under RCRA.

Container disposal:
Dispose of in a licensed facility. Recommend crushing, puncturing or other means to prevent unauthorized use of used containers.

14. Transport Information

Land transport
USDOT
Not classified as a dangerous good under transport regulations

Sea transport
IMDG

Hazard class:	9
Packing group:	III
ID number:	UN 3082
Hazard label:	9, EHSM
Marine pollutant:	YES

Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains RETINOL, SORBITAN, MONOLAURATE, ETHOXYLATED)

Air transport

IATA/ICAO	
Hazard class:	9
Packing group:	III
ID number:	UN 3082
Hazard label:	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains RETINOL, SORBITAN, MONOLAURATE, ETHOXYLATED)

15. Regulatory Information

Federal Regulations

Registration status:

Cosmetic	TSCA, US	released / exempt
Chemical	TSCA, US	released / listed

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

CERCLA RQ	CAS Number	Chemical name
100 LBS	64-17-5; 109-89-7	Ethanol; diethylamine

State regulations

State RTK	CAS Number	Chemical name
NJ	128-37-0	BHT
	25013-16-5	BHA
PA	128-37-0	BHT

Safe Drinking Water & Toxic Enforcement Act, CA Prop. 65:

WARNING: This product can expose you to chemicals including ETHANOL IN ALCOHOLIC BEVERAGES, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.

NFPA Hazard codes:
Health: 2 Fire: 1 Reactivity: 0 Special:

16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2025/12/04

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END OF DATA SHEET